

[21]**SARDAR PATEL UNIVERSITY**

External Examination (CBCS)

B. Sc. (CA & IT) (NC)

M. Sc. IT (Integrated) – 1st Semester

PS01CIIT03: Problem solving methodology and programming in 'C'

22nd February, Monday– 2021

Time: 10:00 am to 12:00 pm

Total Marks: 70

Q-1 Select an appropriate option.

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1. _____ is a step by step approach to solve any problem.
(a) Process (b) Programming Language (c) Algorithm (d) Compiler
2. _____ is a pictorial representation of an algorithm.
(a) Data Diagram (b) Flow Chat (c) Pie Chart (d) Program
3. What is the standard decision symbol for a flowchart?
(a) Circle (b) Lozenge (c) Diamond (d) Square
4. The value can be changed during program execution is known as _____.
(a) Variable (b) constant (c) Operator (d) None of these
5. The combination of ? and : is known as _____ operator.
(a) Ternary (b) Arithmetic (c) dot (d) Relational
6. _____ statement skips the remaining statement of the loop and control transfer to the condition.
(a) Break (b) Continue (c) If (d) Switch
7. For Initialization of array list of value separated by _____.
(a) Question marks (?) (b) Commas (,)
(c) Exclamatory marks (!) (d) None of these
8. '\0' is _____.
(a) Null Character (b) Character Value
(c) Escape Sequence (d) Symbolic Constant
9. Ampersand (&) is not required when read the _____ value.
(a) Int (b) Float (c) String (d) None of these
10. _____ function is use to read the string.
(a) scanf() (b) printf() (c) strrev() (d) None of these

Q-2 Fill in the blanks / True or False

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1. Assembly language is easily re-locatable language. (True/False)
2. C language is high level language. (True/False)

3. The range of int data type is 4 bytes. (True/False)
4. %f format specifier used for integer data type. (True/False)
5. For declaring one- dimensional array _____ subscript is use.
6. _____ loop execute at least once.
7. The _____ string function is use to joins two string.
8. strcmp () function is use for _____ two string.

Q-3 Answer the following questions. (Attempt any **TEN**)

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1. What is an algorithm? List Characteristics of an algorithm.
2. What is an Editor? Give two examples of it.
3. What is Translator?
4. Draw the basic structure of C program.
5. Write rules of variable name.
6. Write difference between pre-increment and post- increment operator.
7. Explain the isalpha() function.
8. What is structure programming? Explain.
9. Explain the isupper() function.
10. Define function? Explain type of function use in C with example.
11. What is user – define function? Give one example of user define function.
12. Write the advantages of function.

Q-4 Answer the following question. (Attempt any **FOUR**)

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1. What is Flowchart? Explain rules to draw flowchart. Also explain symbols used to draw flowchart.
2. Write an algorithm for the followings:
 - (i) To find out minimum from N numbers.
 - (ii) To check whether inputted number is prime or not.
3. What is operator? Explain different operators with example.
4. Explain if..else & nested if statement with syntax and example.
5. What is an array? Explain declaration of 1D and 2D array in C with syntax and example.
6. Explain break and continue statements in C with examples.
7. Explain the following function category with example:
 - (i) with argument and no return value
 - (ii) with argument and return value
8. Explain the following library function with syntax and example:
 - (i) strcmp() (ii) strrev() (iii) strcat() (iv) strlen()

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